



Cryptographic Resistance of the Simple and Generalized Logistic Map with Coupling

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This study compares the cryptographic resistance of two chaos-based systems: the simple coupled logistic map and the generalized coupled logistic map. Their security was evaluated through differential and linear cryptanalysis, and a chaos-based attack was performed via parameter estimation. For the latter, a mean squared error function —previously analyzed to delimit valid attack methods— was optimized using the Chaos Behaved Particle Swarm Optimization (CBPSO) algorithm. The results revealed a significant security gap in the simple coupled logistic map regarding the chaotic attack. Conversely, the generalized coupled logistic map proved to be robust against statistical and chaotic attacks, confirming its superior security for cryptographic applications.

1. Introduction

Due to technological advances and the development of post-quantum cryptography, chaotic cryptography has acquired notable attention for the intrinsic entropy it provides, constituting an alternative to the computational difficulty of primes on which classical algorithms are based. Beyond the post-quantum realm, its relevance has grown substantially in applications such as medical image encryption, increasing academic interest in chaotic systems to generate secure algorithms. However, these same systems present important flaws such as dynamic degradation, high encryption-decryption times, and vulnerabilities specific to each implementation, this last aspect being the central focus of the present work.

Historically, numerous investigations on chaotic encryption have focused on resilience against conventional attacks or statistical arguments to demonstrate security. However, this approach has proven insufficient by not adequately considering attacks that exploit system identification and its fundamental nature. This limitation has led to algorithms presented at specialized conferences revealing security gaps shortly after publication [1], justifying the focus on attacks based specifically on chaos.

This work evaluates the strength of the simple [2] and generalized [3] coupled logistic map, both proposed

by Hugo Solís, against attacks that seek to obtain the chaotic map points to determine the generation keys. Beyond a conventional empirical analysis, the intrinsic nature of the map is studied to establish its advantages in cryptographic implementations, analyzing the difficulty of recovering keys with unknown initial conditions through methods such as CBPSO [4]. In parallel, a comparative differential and linear cryptanalysis is performed between systems based on the simple and generalized coupled logistic maps, establishing a quantitative contrast of their cryptographic resistance. In summary, the goal is to determine the real security of the 2 mentioned logistic maps against attacks of a chaotic nature and traditional cryptographic analyses.

2. Objectives

2.1. General Objectives

To determine the security of the coupled and generalized logistic map against differential type and chaotic attacks.

2.2. Specific Objectives

1. Analyze the error surface of both logistic maps to identify properties that hinder optimization and by extension numerical methods in a chaotic type attack.

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2. Implement CBPSO to estimate parameters without knowledge of the initial points for the chaos-based attack in both systems.
3. Evaluate resistance against differential and linear attacks by means of the coupled logistic map and the generalized one with coupling.
4. Compare the cryptographic resistance between both systems based on logistic maps.
5. Determine the general security of systems with chaotic algorithms based on logistic maps.

3. Logistic Maps with Coupling

3.1. Simple Coupled Logistic Map

Principally, it possesses a wide range of variables that provoke chaotic behavior, which contributes to a more secure algorithm. Furthermore, its low dimensionality makes it more efficient compared to multidimensional maps that have been attempted for use in chaotic cryptography.

$$f(x) = \mu x(1 - x) \quad (1)$$

Where $0 \leq \mu \leq 4$

Then, a linear coupling is performed as follows, [2]:

$$x_{n+1} = f_1(x_n) + \alpha(y_n - x_n) \quad (2)$$

$$y_{n+1} = f_2(y_n) - \alpha(x_n - y_n) \quad (3)$$

With $f_k(x)$ being the map and α a coupling parameter limited between $0 \leq \alpha \leq 1$. The symmetric case happens when $f_1(x) = f_2(x)$, but many of the vulnerabilities of chaotic maps as encryption algorithms are related to symmetric algorithms [1]. For this very reason, there is interest in the asymmetry of this algorithm and the coupling, which adds multiple factors of additional complexity to the system. Furthermore, it is relevant to note how x_n corresponds to the sent signal, while y_n corresponds to the received one. Therefore, the attack to be performed assumes a possible interception of a communication channel such that the attacker has the capacity to send and receive a message or a fraction thereof. Finally, the coupled logistic map for cryptographic use is obtained by implementing an adjustable parameter A and evaluating for each bit of the message.

$$x_{n+1} = f_1(x_n) + \alpha(y_n - x_n) + A \cdot m(n) \quad (4)$$

$$y_{n+1} = f_2(y_n) - \alpha(y_n - x_n) \quad (5)$$

For the chosen map, each bit is evaluated in 50 iterations of the same.

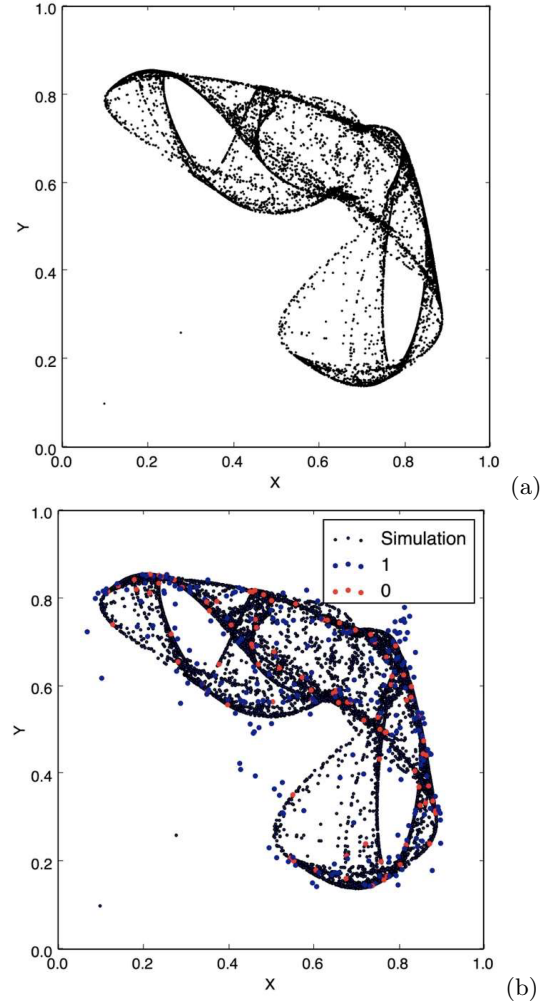


Figure 1: (a) Simple logistic map with coupling and (b) the bit evaluation for its encryption process [2]

3.2. Generalized Coupled Logistic Map

Lawnik [5] originally defines the generalized logistic map, which constitutes a family of maps, and also elaborates on its advantages. Principally, it possesses a wide range of variables that provoke chaotic behavior, which contributes to a more secure algorithm. Furthermore, its low dimensionality makes it more efficient compared to multidimensional maps that have been attempted for use in chaotic cryptography.

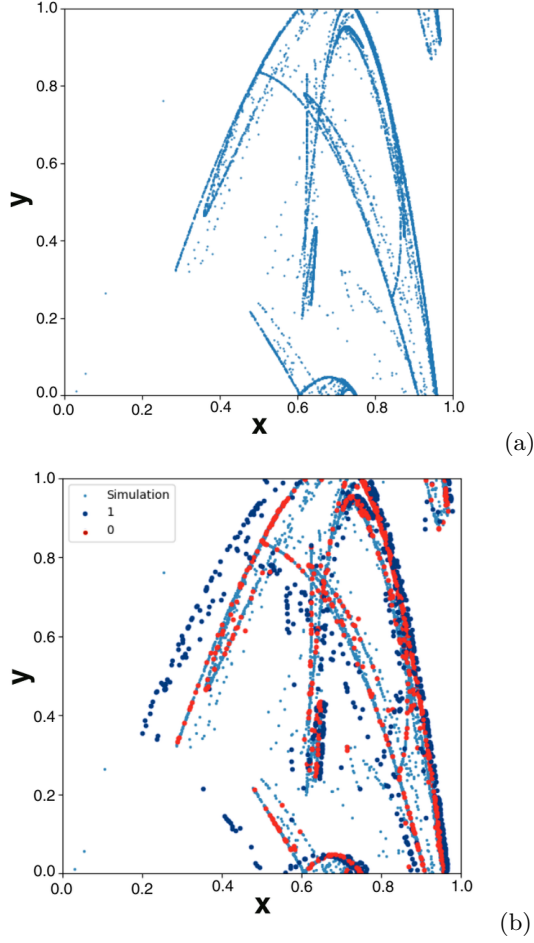


Figure 2: (a) Logistic map with generalized coupling and (b) the bit evaluation for its encryption process [2]

The equation is defined as follows:

$$x_{k+1} = \begin{cases} q \left(1 - \left| \frac{p - x_k}{p} \right|^r \right), & 0 \leq x_k \leq p \\ q \left(1 - \left| \frac{p - x_k}{1 - p} \right|^r \right), & p \leq x_k \leq 1 \end{cases} \quad (6)$$

Where $p \in [0, 1]$, $q \in [0, 1]$ and $r = 2$.

Then, asymmetric coupling is also performed. For this case, one will seek to obtain the correct combination of parameters $\mu, \mu_1, \nu, \nu_1, \alpha$, where $q \rightarrow \nu, p \rightarrow \mu$ is redefined for $f_1(r)$ and $q \rightarrow \nu_1, p \rightarrow \mu_1$ for $f_2(r)$.

The coupling is applied in a similar manner to the simple logistic map case, using the same equations 4 and 5, but with the generalized logistic map instead of the simple one. It will be noted as the other sections are elaborated that both attack methods are implemented

in extremely similar ways for both the coupled logistic map and the generalized logistic map with coupling.

4. Differential and linear cryptanalysis

The main objective of cryptanalysis consists of determining information about the plaintext or the key itself through the systematic examination of the relationships between the inputs and outputs of the algorithm. Over time, cryptanalysis has developed diverse techniques —among them differential analysis and linear analysis— that allow measuring the robustness of a cryptographic scheme against attacks that exploit correlations or internal patterns in its operation. These tools are fundamental to validate the strength and reliability of new encryption systems, including those based on chaotic dynamics.

4.1. Differential cryptanalysis

Differential cryptanalysis is an analysis technique against attacks that seek to find the key in an iterated cipher. This type of attack seeks to analyze the effect of particular differences in an input message on a cryptographic system. These differences can be used to assign probabilities to possible keys and locate the most probable key [6]. The differential approximation probability, DAP , is defined as:

$$DAP = \max_{\Delta x \neq 0, \Delta y} \left(\frac{\#(x \in \Omega \mid f(x) \oplus f(x \oplus \Delta x) = \Delta y)}{2^n} \right) \quad (7)$$

Where x and x^* are messages from a group of messages Ω such that, $x, x^* \in \Omega$, in such a way that $\Delta x = x \oplus x^*$, where $\oplus$ is that operation that obtains the difference between the input messages x and x^* . Furthermore, $y = f(x)$ and $y^* = f(x^*)$, and their difference will be $\Delta y = y \oplus y^*$. Finally, 2^n is the cardinality of the group Ω [6].

This type of cryptanalysis proves particularly effective in symmetric cryptographic systems. Consequently, when applying it to chaotic systems based on logistic maps, the results obtained are expected to be minimal, which would evidence the asymmetric nature of the system.

4.2. Linear cryptanalysis

Linear cryptanalysis is an analysis technique against attacks that seeks to construct an approximate linear

expression of the cipher of the system under study [6]. The non-linearity of the encryption function $f(x)$ is measured using the linear approximation probability, LAP , which is defined as:

$$LAP = \max_{a,b \neq 0} \left(\frac{\#(x \in \Omega \mid x \bullet a = f(x) \bullet b) - 2^{n-1}}{2^{n-1}} \right)^2 \quad (8)$$

Where a and b are two bit masks and $\bullet$ denotes the bit-wise product. Understanding this in depth goes beyond the objective of the application of this probability in the system of study. More information can be obtained in the article by Alvarez and Li [6].

Analogously, this type of cryptanalysis proves especially effective in symmetric cryptographic systems. Therefore, the results obtained are expected to evidence the asymmetric nature of the analyzed system.

4.3. Cryptographic analysis

For the cryptanalysis of both cryptographic systems (the one based on coupled logistic maps and the one based on generalized logistic maps), the results were contrasted considering all possible combinations of the selected list of bits [A,B,0,1]. This list was chosen for including both text-type and numerical inputs, while maintaining a manageable level of complexity. It is recommended to expand the list of inputs if greater processing capacity is available. The approximation probabilities were calculated following the methodology proposed by Álvarez and Li [6], as well as the systematic approach previously implemented by Solís-Sánchez [3].

5. Cryptanalysis results

Upon analyzing the selected list [A,B,0,1] following the definitions established by Álvarez and Li [6], a study was first carried out for 8 bits, considering combinations of one element of the list with another. As expected, in some iterations the difference between the input messages is null, given that the same element is repeated in both positions. In other cases, messages with small or considerable differences are combined. For 8 bits, the results obtained are shown in Table I:

Table I: Maximum values of differential approximation probability (DAP) and linear (LAP) for 8 bits in cryptographic systems.

	Generalized System	Coupled System
Max DAP	0.0234375	0.015625
Max LAP	0.0007244981825	0.0007244981825

As can be observed, the differential approximation probability is relatively low in both systems, with greater security predominating in the system based on coupled logistic maps. In contrast, the linear approximation probability remains identical in both systems.

According to what was demonstrated in previous sections, one would expect the system based on generalized logistic maps to be more secure even under these analyses. However, given that these cryptanalysis methods are predominantly effective in symmetric systems, the results obtained could be considered a false positive.

In the case of linear analysis, the equality of probabilities suggests that neither of the systems presents an average linear advantage over the other. This coincidence could also be interpreted as a false positive, in the sense that subtle differences might not be detectable with the computing capacity used. Nevertheless, the observed equality indicates that, during the encryption and decryption processes, there exists an average linearity decipherable with a relatively low probability.

Next, a study was carried out for 16 bits, using the same list of elements, but combining two of them to form each message, over which iterations were performed. The results are presented in Table II:

Table II: Maximum values of differential approximation probability (DAP) and linear (LAP) for 16 bits in cryptographic systems.

	Generalized System	Coupled System
Max DAP	0.0001678466797	0.000244140625
Max LAP	0.0000000001054283614	0.0000000001054283614

This 16-bit analysis proves particularly revealing compared to the 8-bit one, clarifying the possible interpretation of previous results as a "false positive". In this case, a lower differential approximation probability is observed for the system based on generalized logistic maps.

The discrepancy between the results for 8 and 16 bits is explained by the asymmetric nature of the system: in each iteration, both systems can vary their DAP, showing greater or lesser security in a non-uniform manner. Therefore, differential cryptanalysis, although useful for identifying vulnerability patterns, proves unreliable in

this context for determining which system is inherently more secure against differential attacks. However, it does evidence the true asymmetry of the algorithm, which increases its cryptographic robustness and reduces the effectiveness of the employed cryptanalysis techniques.

Regarding the linear analysis, the approximation probability remains identical for both systems even at 16 bits. This reinforces the existence of non-null linearity in the algorithms, but indicates that neither of the systems presents a significant advantage over the other in this aspect.

The 16-bit calculation on the systems proves particularly demanding due to the extensive computation times required. Empirically, it is observed that the rate of increase in wait times follows an approximate behavior according to the following relationship:

Upon introducing a list of k elements for analysis, the calculation time t (in seconds) can be estimated by the function:

$$t = 32k^{2n} \quad (9)$$

where n represents the number of input elements considered per comparison. For example, for the comparison A + A" (8 bits compared) one has $n = 1$, while for AA + AA" (16 bits compared) one has $n = 2$.

6. Chaos-Based Attack

Within the field of cryptanalysis, in addition to differential and linear analysis, which study the statistical nature of the system in more detail, chaos-based attacks are built more upon prior knowledge of the system's nature, that is, knowledge of the algorithm being worked on. Differential analysis can recover a message without internal knowledge of the encryption operation, unlike the chaos-based attack. In turn, within the different factors that constitute the integral nature of any algorithm, there is dynamic degradation, which determines if a chaotic system exits chaos; in turn, Lyapunov exponents are also studied, which quantify the entropy generated by each system [5]. However, focus will be placed on the ease of estimating the parameters of each algorithm, which can be done with multiple types of mathematical, heuristic, and metaheuristic methods. Therefore, the methods available to recover the algorithm keys are first delimited, and subsequently an attack is performed.

6.1. Objective Function

One seeks to recover the parameters, or a sufficiently precise approximation, of both logistic maps exclusively

from the map points. Therefore, an objective function equivalent to the mean squared error is established.

$$\frac{1}{N} \sum_{i=2}^N \sum_{j=1}^2 (p_{ij,true} - p_{ij,estimated})^2 \quad (10)$$

With N being the number of points on the map, $p_1 = x$ and $p_2 = y$. In this way, for complete recovery of the map parameters, minimizing the MSE is sought. Likewise, in practice, it is important to consider interference noise in the recovered signal in the form of AWGN type Gaussian noise on the received signal, which would add orders of magnitude to the mean squared error [7]; this would take the form of $y_{n,AWGN} = y_n + n$. However, for the terms of this project, the noise-free scenario will be worked on, so that the algorithm's security will be evaluated without taking into consideration external factors that affect the optimization process, equally concluding on its security.

Furthermore, in order to simplify the optimization process described in the following section, optimization will be performed starting from the second point, to be able to solve without needing to search for two additional variables, these being the initial conditions; for this reason, one starts from $i = 2$ instead of $i = 1$. In other words, the first evaluated values of the objective function are x_2, y_2 , which are calculated by means of x_1, y_1 , which are known.

Then, $N = 30$ is established, equal to that used in parameter estimation on the simple logistic map [8], since it is the value of N that provides the best results when minimizing. This occurs because a larger quantity of points generates noise over the optimization process, reducing its precision. Then, because the objective function presents a smaller quantity of points than the 50 iterations required to encrypt a single bit, an attack can be performed with only a fraction of the required points. By starting from an attacker who can send or receive a message, this can be arbitrarily established as $m(n) = 0$ such that the optimization problem is reduced to one that only seeks the correct combination of parameters.

Furthermore, due to the internal operation of both maps with coupling, and the decimal precision of their calculations, it is established that a map estimated with an error function with orders of magnitude equal to or less than 10^{-6} is indistinguishable from the original. Therefore, an optimization process that obtains results of that magnitude would demonstrate a security gap in the algorithm.

6.2. Selection of the optimization method

First, it is briefly mentioned that, due to the coupling and the described asymmetry of both logistic maps, they are particularly secure against brute force attacks. The computational complexity of this type of attack is $O(n^k)$, with n being the number of variables and $k = \frac{1}{h} + 1$ the number of possible combinations that exist, which in this case depend on the step size h , or the decimal precision, with which the search is being performed within the space $[0,1]$.

Then, three types of methods for approximating functions are considered: gradient, iterative, and mixed. Specifically, Nelder-Mead, an iterative method that generates a polygon called a simplex to optimize, and Levenberg-Marquadt, which performs a combination of an iterative method with the steepest descent method, stand out among two well-studied and used tools in the field of chaotic systems, together with the more general gradient method [9]. In the case where one has a non-smooth objective function, and by extension non-differentiable, this allows discarding any gradient method, which uses Jacobians like Levenberg-Marquadt or any other form of derivative.

Therefore, direct search methods are required, such as Nelder-Mead. However, said method, along with other analogues, tends to easily converge in local minima. Then, if a high amount of local minima and non-smooth behavior is observed in the MSE, then it would be determined that gradient, iterative, and mixed methods like those previously mentioned are not valid for performing an attack.

6.3. Metaheuristic Algorithms, PSO and CBPSO

Metaheuristic algorithms are those combinations of heuristic methods, or based on approximations, with the objective of solving a problem or a set of problems that cannot be obtained efficiently, or at all, with classical methods. Commonly they are used in optimization problems like parameter estimation, when there is little information about the system. In these cases, speed for obtaining the solution is prioritized over its precision, and therefore it is a valid alternative to brute force attacks. Currently, the family of most investigated metaheuristic algorithms are Differential Evolution (DE) and Particle Swarm Optimization (PSO).

Even if DE tends to present better average performance across different industries, PSO possesses some advantages in the studied case, which is parameter estimation. It is emphasized that traditional PSO and those

variants close to it generally do not have good performance for cases of greater complexity; however, there is a greater quantity of variants for specific cases [10]. Additionally, there is no notable interest in research applying DE for parameter estimation compared to PSO, which has multiple variants created with the specific purpose of approximating chaotic functions and recovering their parameters. Therefore, PSO is chosen with an appropriate variant, this being CBPSO [11].

6.3.1. PSO

Regular PSO establishes a set of candidate solutions of a system as a swarm of particles with position, velocity, and a search algorithm.

$$X_i(t+1) = X_i(t) + V_i(t+1) \quad (11)$$

$$V_i(t+1) = wV_i + c_1r(X_{bi}(t) - X_i(t)) + c_2r(X_{bg}(t) - X_i(t)) \quad (12)$$

Where X_{bi} and X_{bg} are the best individual solution of particle i and the best global position respectively. Furthermore, t is the iteration number and r a number generated randomly between 0 and 1. Then, the main expressions describing the movement w , c_1 and c_2 are presented, being these the inertia weight, the cognitive constant, and the social constant respectively. Said another way, there is the tendency of the candidate solution to maintain the direction it follows, which is influenced by its own experience and that of nearby solutions [12].

6.3.2. CBPSO

Specifically, Chaos Behaved Particle Swarm Optimization, or "Chaos Behaved Particle Swarm Optimization" implements a series of changes that improve its performance against optimizations related to chaotic algorithms, as described by Yuexi Peng et al [4].

As previously mentioned, traditional PSO is not as efficient nor precise unless adapted to the problem worked on. For example, it is better to work the inertia weight as a function of iterations, which has already been done for estimations of chaotic algorithms, like the non-generalized logistic map [8]. While, on average, more precise results are obtained by making w a linear function of iterations, for chaotic algorithms one seeks to approach as quickly as possible the approximate area of the global minimum, in order to avoid getting trapped in local minima; then, one seeks for the particle velocity to decrease in order to better adjust its position to the precise solution. Therefore, an exponential function of

iterations is chosen for the inertia weight.

$$w(t) = (w_{max} - w_{min})e^{-p(p/T)} + w_{min} \quad (13)$$

Furthermore, chaotic initialization is implemented instead of random initialization, using the logistic map.

$$x(n+1) = \mu x(n)[1 - x(n)], n = 0, 1, 2, \dots \quad (14)$$

$$X(n+1) = X_{min} + (X_{max} - X_{min})x(n) \quad (15)$$

With $x(n)$ being the system variable, $x \in (0, 1)$, but $x(0) \notin \{0.25, 0.5, 0.75\}$. Then, the particle updates its position according to the chaotic search:

$$X(n) = [X_{bg}(t) - a] + \{[X_{bg}(t) - aX_{bg}(t)]\}x(n) \quad (16)$$

Where $0 < a < 1$

Finally, for the optimization process the following is established: $p = 100, T = 250, N_p = 1000$. The variable p is chosen according to what is stated in the original CBPSO method; in the case of T , it is noted that starting from 250 no sufficiently relevant improvements in results are found to justify the high convergence times; for the number of candidate solutions, or particles N_p , a modification was made on what is classically performed in PSO and its variants. Generally, one works $30 \leq N_p \leq 50$, however there are scenarios where results improve substantially when more particles are implemented [13], mainly when there are equations with complex structures like those of maps with coupling. By covering a greater quantity of the problem space, the possibility of converging in a local minimum is minimized.

6.4. Cryptographic analysis

The computer used consists of an Intel Core i5-7400 processor and contains 16 GB of RAM. The entire process of map generation, statement of the objective function, and its optimization via CBPSO is programmed in Jupyter Notebook 7.3, exclusively using Python v3.12.5.

For both maps (simple and generalized), different graphs related to an objective function were made, this being the mean squared error (MSE) between the original map and the one obtained via estimated parameters, both evaluated for N points. With this, three types of graphs were obtained for each parameter separately, taking the other keys as known: surface with a known initial value y_0 , contour with a known initial value y_0 , and 2D with both initial values known. It is noted that the graphs do not vary in their nature if the known value is x_0 or y_0 . The graphs for each parameter presented the same nature. The results of this section delimited the methods available to perform an attack.

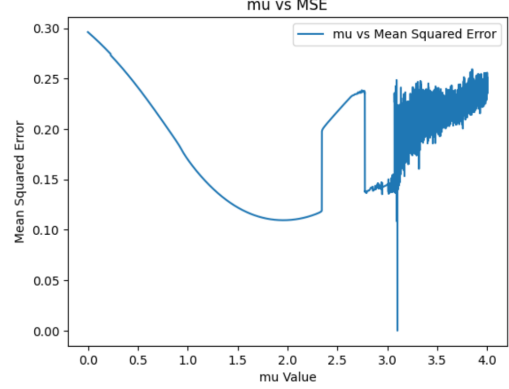


Figure 3: MSE function with respect to μ for known initial conditions x_0 & y_0 for the simple coupled logistic map

Then, the chaotic map was generated with N points x and y , with which the objective function is defined. It is established that the map possesses a security vulnerability if this is minimized to an order of magnitude less than 10^{-6} . The CBPSO algorithm was programmed, which, after 100 different optimization cycles, compiled the following information: the average MSE, average time per cycle, with their respective standard deviations, the cycles with best results, and the graph of the original map points compared with those of the map using the parameters obtained after the optimization process.

6.5. Parameter estimation results

6.5.1. Objective function analysis for both maps

The analysis of the objective function revealed similar characteristics in both systems. For the coupled logistic map, a non-smooth and non-differentiable structure was observed, along with a wide quantity of local minima scattered throughout the solution space. Specifically for parameter μ of the coupled map, a prominent local minimum is observed in the approximate range of 0.5; this class of structures hinders the optimization process for most solution methods.

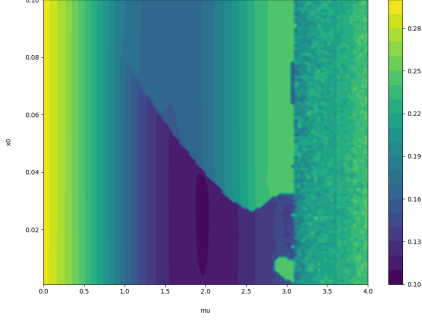


Figure 4: Contour graph with respect to μ for known x_0

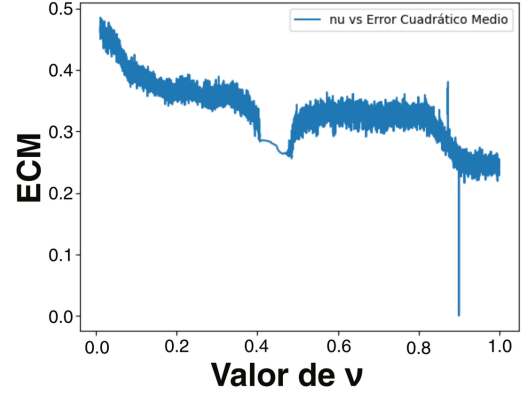


Figure 6: MSE function with respect to ν for known initial conditions x_0 & y_0 (generalized map)

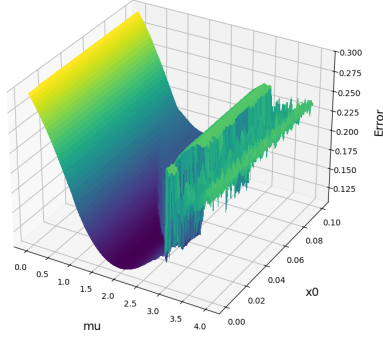


Figure 5: Surface graph with respect to μ for known x_0

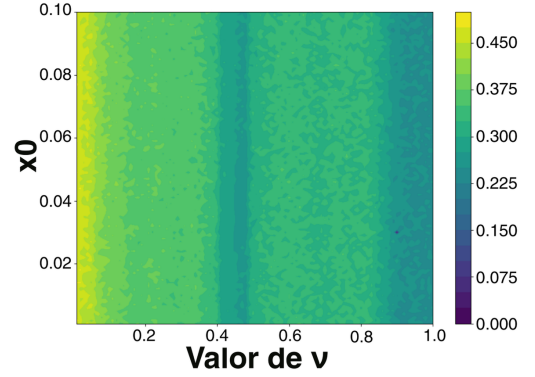


Figure 7: Contour graph with respect to ν for known x_0

For the generalized logistic map, the behavior was similar throughout the different parameters. In all cases a non-smooth structure was found, and thus non-differentiable, along with a wide quantity of local minima. The case for ν is specifically shown, since it best illustrates the described nature of the parameters.

In both systems, no pattern close to the region where the global minimum solution is found is recovered, which contributes to the cryptographic security of the algorithms. The contour and surface graphs show the same behavior in three dimensions for both maps. This common characteristic invalidates iterative methods that use derivatives or mixed methods, justifying the choice of metaheuristic algorithms for the attack.

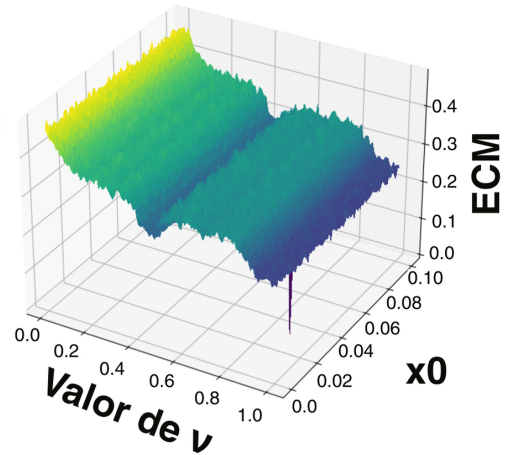


Figure 8: Surface graph with respect to ν for known x_0

6.6. Attack with CBPSO: Comparative results

The results of the attack with CBPSO showed significant differences between both systems.

6.6.1. Simple Coupled Logistic Map

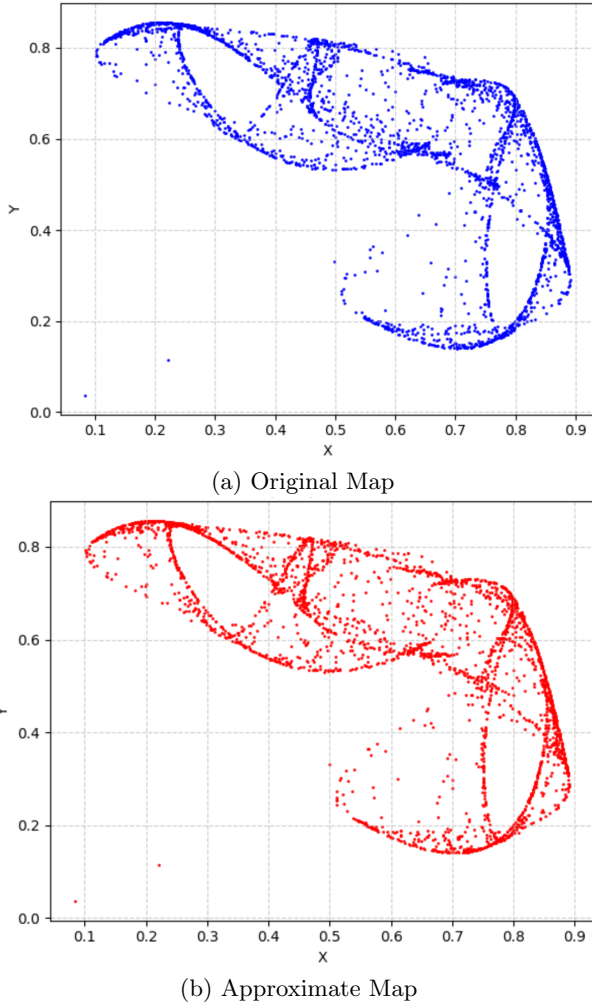


Figure 9: Comparison between (a) the original map and (b) the approximate one with the parameters obtained from the best CBPSO result for the coupled map

For the coupled map, the results were notably successful. Throughout all processes, the mean squared error objective function reported solutions within the range $[3.19 \times 10^{-10}, 3.28 \times 10^{-11}]$ with an average of $5.08 \times 10^{-11} \pm 3.32 \times 10^{-12}$. Each cycle had an average duration of $27.46 \pm 5.49s$.

Iteration	μ	μ_1	ν	ν_1	α	MSE
Original	0.9100	0.8500	0.9000	0.3330	0.3314	0
Best result	0.8413	0.8331	0.8337	0.2963	0.3205	0.0028
Difference %	7.55%	1.99%	7.36%	11.03%	3.29%	0.28

Table IV: Summary of the best result of the MSE objective function for 100 CBPSO cycles (generalized map)

Table III: Summary of the best result of the MSE objective function for 100 CBPSO cycles (coupled map). The MSE was 3.19×10^{-19} .

Iteration	μ	μ_1	α	x_0	y_0
Original	3.10000	2.90000	0.3314	0.03000	0.01000
Best result	3.09999	2.90000	0.33139	0.03000	0.00999
Difference %	0.0003%	0.0000%	0.0030%	0.0000%	0.1000%

Even the least precise results were multiple orders of magnitude above the sought precision of 10^{-6} . The parameters had a negligible average difference with respect to the originals, which implies that the original parameters were successfully recovered. In figure 9 it is observed that, with an intercepted fraction of the original map, obtaining the parameters resulted in a map indistinguishable from the sought one.

6.6.2. Generalized Coupled Logistic Map

For the generalized map, the results were considerably different. Throughout all processes, the mean squared error objective function reported solutions within the range $[0.0360, 0.0028]$ with an average of 0.0165 ± 0.0092 . Each cycle had an average duration of $31.61 \pm 1.13s$.

The most precise results were obtained 31% of the time, and happened when an order of magnitude of 10^{-3} was reached, which does not correspond to the objective of 10^{-6} . The parameters had an average of 6.24% difference with respect to the originals, which implies that the original parameters were not successfully recovered. In figure 10 it is observed that, even if a fraction of the original map structure is recovered, the difference in parameters resulted in a map distinguishable from the sought one.

It is also relevant to mention that, throughout the 100 cycles, there were no patterns regarding the values that the optimized parameters took. This means that local minima like the one shown in figure 6 were successfully avoided, which is due to the exponential inertia weight, along with the chaotic initialization and search; otherwise, the local minimum of this or similarly for any of the other parameters would have been constantly converged upon. This is a positive factor regarding the algorithm's

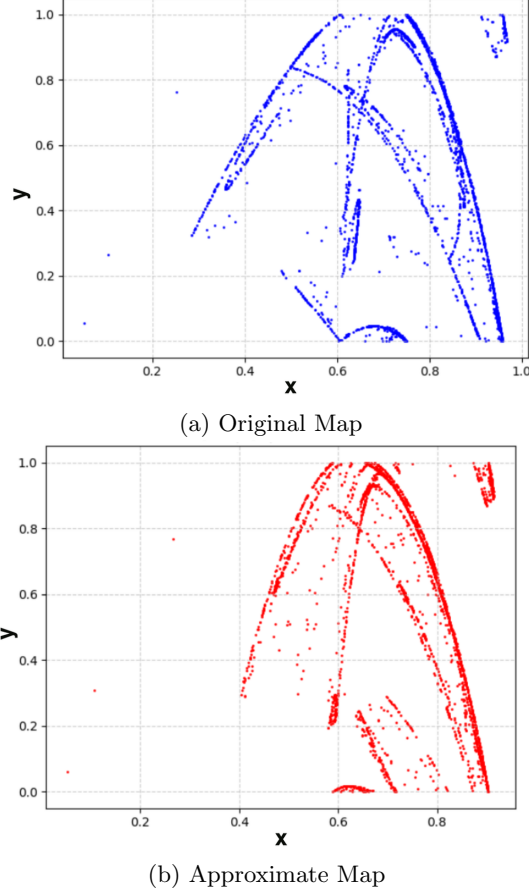


Figure 10: Comparison between the original map and the recovered one for the best CBPSO result of the generalized map

security due to the implication of an approximately uniform difficulty regarding the ease of obtaining any of the parameters.

6.6.3. Optimization progress

In both cases, the algorithm's progress attenuated after a certain number of iterations, corroborating what was established regarding $T = 250$ being an appropriate quantity of iterations to optimize. For the coupled map, convergence stabilized around 25 iterations, while for the generalized map this occurred around 150 iterations. This means that, besides possessing a security gap, said gap can be obtained in few iterations, and therefore short times. On the other hand, the attenuation of progress for the generalized coupled logistic map is sufficient that it is not possible to increase the number of

iterations such that a guaranteed solution is obtained in a reasonable time, demonstrating its security.

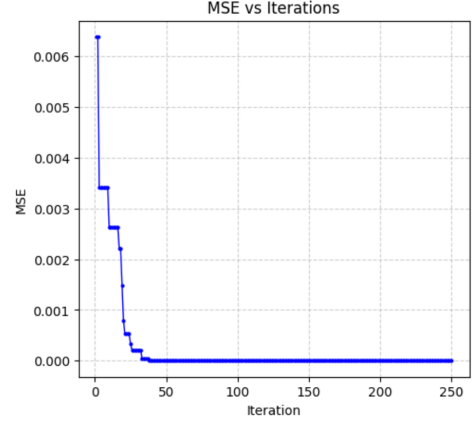


Figure 11: Change of MSE during the optimization process for CBPSO for the coupled logistic map

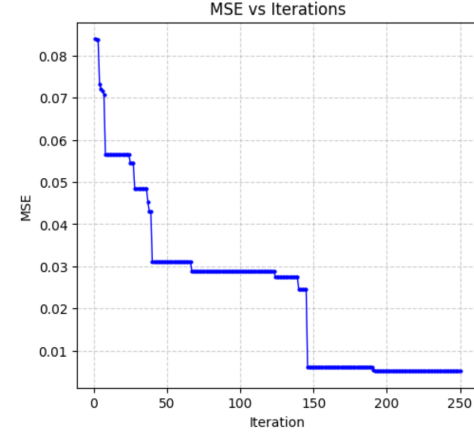


Figure 12: Change of MSE during the optimization process for CBPSO for the generalized logistic map

7. Conclusions

The strength of both logistic maps with coupling against different types of attacks was studied in a comparative manner. Specifically, the capacity to regenerate the maps from the points was investigated, establishing an objective function equivalent to the mean squared error and studying the difficulty of estimating the parameters from it. Then, optimization was performed with CBPSO, a metaheuristic method, and multiple factors of the algorithms were determined.

For the coupled logistic map, a significant vulnerability was found against chaos-based attacks. The greatest strength of this map presents a security gap regarding chaos-based attacks given a fraction of 30 points of the generated map, where one has the optimal quantity of points to perform the attack. Furthermore, the original parameters and initial conditions could be recovered with minimal difference, such that a precision of $\times 10^{-19}$ was obtained in the approximate map points with respect to the original, so that they are indistinguishable. Then, even if additive Gaussian noise (AWGN) was not added, which could improve its security, the results were equally sufficiently conclusive that an absolute security gap can be determined.

For the generalized logistic map, the results were considerably different. The attack with CBPSO did not manage to recover the parameters with the necessary precision, obtaining average differences of 6.24% in the parameters and a minimum MSE of 0.0028, well above the security threshold of 10^{-6} . Even if metaheuristic algorithms have the potential to recover the generalized logistic map, and manage to approximate with precision of up to 1.99% in some parameters, current methods are still not capable of fully recovering a signal sent by it.

Regarding the differential cryptanalysis results, these indicate that the system based on the generalized logistic map offers superior robustness and greater resistance against differential attacks compared to the coupled system. However, the persistence of a non-null and constant linear approximation probability in both systems evidences an intrinsic vulnerability that neither of the algorithms manages to eliminate, highlighting the need for complementary or alternative approaches to guarantee complete cryptographic security.

Possible improvements for both systems include considering making A a function that contributes to the system's chaos, but the stated attack approach is not affected by said variable and furthermore, it would notably increase encryption times. Then, for the coupled map, using a generalized version or a variant with a greater number of parameters to optimize, such that it presents a greater number of parameters to optimize, could endow it with greater security. For the generalized map, the addition of Gaussian noise (AWGN) is a viable option so that the algorithm remains effective against attacks on chaos as methods evolve.

If greater computational capacity or more optimized code is available, it is recommended to expand the analysis using longer lists and evaluating systems with a greater number of bits, in order to obtain a more precise characterization of the cryptographic resistance of the algorithms.

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A. Code repository

The programs used for the described procedure are attached in [Github](#), along with additional material such as the rest of the two-dimensional, surface, and contour graphs for the parameters in both the simple coupled logistic map and the generalized coupled logistic map.